Three Fidelities: Evaluating LLM Twins Against Behavioural Message-Testing Trials

Effect, distributional and mechanism fidelity on four open randomised experiments, with a prompt-perturbation audit

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Abstract

Commercial "synthetic respondent" tools predict how an audience will react to a public message by prompting language-model personas. The published evidence for such tools rests almost entirely on survey items, and almost entirely on one number per model from one prompt. We propose an evaluation that separates three things a simulator can be faithful to, and test it on four open randomised experiments with behavioural or quasi-behavioural outcomes and individual-level demographics: snack warning labels (Clarke et al. 2020, UK, N=4,134), salt warning labels (Evans et al. 2025, UK, N=2,391), vaccine-information framings with a null human effect (Kerr et al. 2021, UK, N=2,488) and front-of-pack label systems (Mora-Plazas et al. 2024, Colombia, N=8,004). Each experiment is replicated with 150 twins: personas generated from real respondents' own covariates, placed in every arm, making the study's decision. We score (i) effect fidelity, whether arms differ as they did for humans, against two human-only reference lines; (ii) distributional fidelity, whether the response distribution per arm matches; (iii) mechanism fidelity, whether the same covariates and covariate×arm interactions predict the outcome; and (iv) individual fidelity against a leave-one-out lookup table. Claude Sonnet 5.5 recovers arm orderings where humans have them (Clarke pooled ρ=0.94) and correctly returns nothing on the null task, but its effect sizes are five times human size on Clarke, it misses the Evans effect entirely, and every label system saturates on Mora-Plazas. Simulated variance is 26–64% of human variance on every dataset and model. Covariate main effects are recovered with the right signs (r=0.84 on Kerr) but at 3–6× human strength, and covariate×arm interactions, the quantity a segment-targeting tool reports, show no agreement anywhere. Twins never beat a lookup table at predicting the individual they were built from. Two further audits qualify every number: asking the same personas which label would move them predicts the reverse of what people did, and 25 semantically equivalent rewordings move a cheap model's score across a range of 0.77 and a frontier model's across 0.39. We release the eval specs and argue that audience simulators should replicate decisions rather than poll opinions, report rankings with a human-calibrated scale rather than levels, publish a null task and a wording family with every score, and not report segment-specific claims at all until interaction fidelity is demonstrated.

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1 Introduction

At least eight funded companies sell LLM "digital twins" that forecast how population segments respond to marketing and public-health messages [5]. The academic evidence is mixed. Conditioned on rich backstories, LLMs reproduce group-level survey distributions reasonably well [1, 12], and aggregated LLM forecasts of survey-experiment treatment effects correlate at $r \approx 0.85$ with realised effects [7]. The same models flatten within-group variance, drift toward socially desirable answers, inflate between-group gaps two- to four-fold and are sensitive to cosmetic prompt changes [17, 2, 13, 3]. Existing benchmarks score survey items [14, 15, 16, 8]; none scores what people did after exposure to a message, none reports a human noise ceiling, and none reports how much a score moves under rewording.

"Similar to a human" is not one property. A simulator can reproduce how a treatment shifts behaviour on average (effect fidelity), the spread of behaviour within a group (distributional fidelity), or the way individual characteristics drive behaviour and moderate treatment (mechanism fidelity; we measure a structural proxy, since true mechanism requires interpretability access). A message-testing product needs the first, is unsafe without the second, and sells the third. This paper builds an evaluation that scores all three against open behavioural trials and reports what two current models achieve.

2 Method

Datasets. Table 1. Selection criteria: randomised exposure to three or more message or label variants, a choice or behavioural outcome, downloadable individual-level data with demographics. Kerr et al. is included as a null task: five information framings that had no detectable effect on vaccination willingness in 2,488 UK adults. A simulator that produces an effect there is manufacturing one.

Table 1: Datasets. Ceiling = split-half Spearman correlation between segment-level arm rankings computed on two random halves of the human sample (gender×age-band segments). Pooled baseline = correlation obtained by predicting every segment with the whole-population ranking.

Study	Country	Arms	N	Outcome	Ceiling	Pooled baseline
Clarke et al. 2020, S2 [4]	UK	6 snack-labelling schemes	4,134	chose lower-energy snack (12 options)	0.61	0.82
Evans & Robinson 2025 [6]	UK	4 salt-warning designs + QR control	2,391	total salt (g) over 6 menu choices	0.42	0.68
Kerr et al. 2021, S1 [9]	UK	5 vaccine-information texts (null)	2,488	Oxford willingness scale, 1–5	0.08	0.28
Mora-Plazas et al. 2024 [10]	Colombia	4 front-of-pack label systems	8,004	chose no-added-sugar drink	0.87	0.90

Twins. For each dataset we sample 150 respondents across arms. Each becomes a twin: a ∼200-word first-person interview transcript generated by the model under test, constrained to the respondent's recorded covariates (age, gender, ethnicity, education, income, BMI, relevant habits) and forbidden from contradicting them. Twins therefore have exactly the covariate composition of a human sample and can be matched back to a real person.

Replication, not polling. Each twin is placed in one arm at a time and makes the study's decision with the study's options: picks one of twelve snacks (Clarke), one item from each of six menus with the arm's symbol printed beside high-salt items (Evans), answers the seven Oxford willingness items after reading the arm's text (Kerr), or chooses between two drinks whose fronts carry the arm's label (Mora-Plazas). Every twin is run in every arm, so arm comparisons are within-composition. Prompts contain no honesty admonitions, for reasons given in Section 3.6.

Scores. Effect: Spearman correlation between human and simulated arm means, pooled and within segment; slope of simulated on human arm effects relative to control (1 = calibrated); top-1 arm; correlation and slope of between-segment differences. Distribution: Wasserstein-1 per arm in outcome units; total variation for discrete outcomes; dispersion ratio (simulated SD / human SD within arm); mean bias over segment×arm cells. Mechanism: OLS of outcome on standardised covariates and arm dummies in each table; correlation and sign agreement of covariate coefficients; the same for arm×covariate interactions; share of outcome variance explained by covariates alone. Individual: twin outcome in its real person's own arm versus that person's outcome, against a leave-one-out (segment, arm) cell mean from the full human sample.

Models and cost. Claude Sonnet 5.5 ($2/$10 per Mtok) and Claude Haiku 5.5 ($0.10/$0.50), temperature 1, one draw per twin per arm. The four-dataset run cost $17.90. Two earlier audits on Clarke used synthetic personas: a design comparison (poll versus replicate, Section 3.5) and a prompt-perturbation study (Section 3.6).

3 Results

3.1 Effect fidelity: ordering yes, magnitude no

Figure 1 and Table 2. On Clarke both models reproduce the human ordering of the six schemes (pooled ρ=0.94) and Sonnet's segment-level rankings (0.77) exceed the human ceiling (0.61), approaching the pooled baseline (0.82). But Sonnet's effect sizes are 5.3× human: the labels explain 55% of simulated choice variance against 2.4% for humans, and twins run from 7% to 99% healthy choice where humans move from 41% to 63%. Haiku's slope of 0.62 looks better calibrated only because its twins barely respond to anything (4% to 17%).

On Evans, neither model recovers the salt effect (pooled ρ=0.10, slope negative). The twins reacted to the label: the share of labelled items chosen fell from 27% under the QR control to 5% under any warning (humans: 48% to 33%). They then chose low-salt unlabelled items, so total salt barely moved. Twins avoid the marked item and ignore the nutrient; humans do a little of both. Top-1 accuracy of 0.67–0.83 here is a tie artefact: the four designs differ by under 0.5 g for humans.

On the null task, Sonnet correctly produces no effect (slope −0.03; arm R^2 0.000). Haiku produces a negative effect of information (slope −0.39, pooled $\rho = -0.60$). The null task separated the two models when the positive tasks did not.

On Mora-Plazas, with back-of-pack sugar figures mentioned in the prompt, every twin on both models chose the healthier drink in every arm. Removing the figures restores the ordering (pooled ρ=0.77) but any label at all saturates to 100% healthy choice, while humans distinguish Nutri-Score (73%) from a nutrient warning (83%).

3.2 Distributional fidelity: collapsed everywhere

Dispersion ratios are 0.26–0.64 on every dataset and model (Figure 2, left). No cell reaches 0.7. Wasserstein distances are large in outcome units: 2.4–2.9 g of salt against a human SD of 3.1 g; 0.45–0.52 scale points on Kerr. Bias is toward the healthier or more compliant answer on three of four datasets. Because twins carry real covariates, this collapse cannot be attributed to synthetic personas; it is the flattening

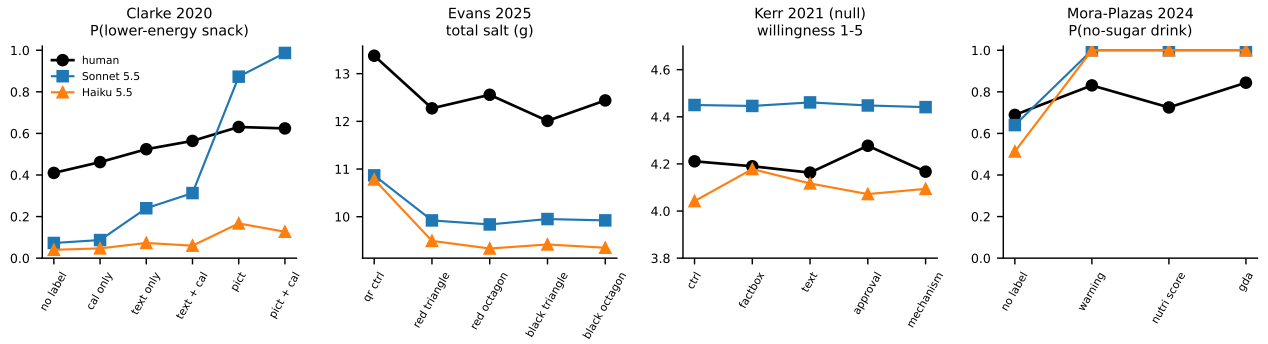


Figure 1: Per-arm outcome, humans (black) versus 150 twins per model. Sonnet recovers the ordering on Clarke and Mora-Plazas with exaggerated range, is flat on Evans, and correctly flat on the null task; Haiku is flat everywhere except Mora-Plazas.

Table 2: Headline scores. Effect slope: simulated arm effects regressed on human arm effects (1 = calibrated). Dispersion: simulated/human SD within arm. Cov. R^2: outcome variance explained by covariates alone, human versus twins. Interaction r: agreement of arm×covariate coefficients. Individual: twin versus own person, against leave-one-out lookup.

Dataset	Model	pooled ρ	seg. ρ	effect slope	dispersion	bias	cov. R^2 h/sim	interaction r	individual (sim vs lookup)
Clarke	Sonnet	0.94	0.77	5.32	0.64	−0.10	0.03 / 0.07	0.03	acc 0.58 vs 0.61
Clarke	Haiku	0.94	0.66	0.62	0.55	−0.45	0.03 / 0.19	−0.02	acc 0.43 vs 0.61
Evans	Sonnet	0.10	0.12	−0.17	0.35	−2.5 g	0.08 / 0.32	−0.57	MAE 3.1 vs 2.2 g
Evans	Haiku	0.10	0.33	−0.19	0.47	−2.9 g	0.08 / 0.12	0.60	MAE 3.7 vs 2.2 g
Kerr (null)	Sonnet	−0.10	−0.26	−0.03	0.40	+0.22	0.10 / 0.59	−0.11	MAE 0.77 vs 0.78
Kerr (null)	Haiku	−0.60	−0.22	−0.39	0.45	−0.13	0.10 / 0.33	0.24	MAE 0.84 vs 0.78
Mora-Plazas	Sonnet	0.77	0.69	0.00	0.26	+0.15	0.00 / 0.01	0.13	acc 0.77 vs 0.79
Mora-Plazas	Haiku	0.77	0.69	0.00	0.27	+0.11	0.00 / 0.00	0.26	acc 0.74 vs 0.79

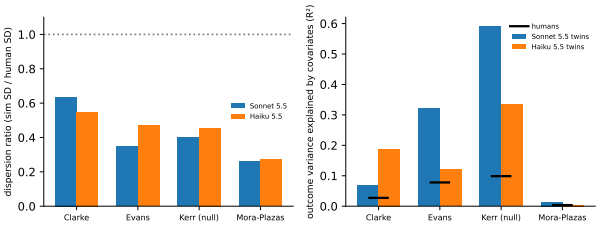


Figure 2: Left: within-arm dispersion ratio; 1 would match human variance. Right: share of outcome variance explained by covariates alone, humans (black ticks) versus twins. Twins are their demographics; humans mostly are not.

reported for survey items [17, 11] reproduced on behavioural outcomes with matched composition.

3.3 Mechanism fidelity: the right levers, pulled too hard; no interactions

Where humans have covariate structure, Sonnet finds it with the right signs: Kerr main-effect correlation 0.84 with 100% sign agreement (politics, age and subjective social status predict willingness the same way); Clarke 0.80 with 83% agreement. Where humans have none (Evans, Mora-Plazas), Sonnet returns none, which is correct. But covariates explain 3–6× more of twin behaviour than of human behaviour (Figure 2, right): Kerr 0.59 versus 0.10, Evans 0.32 versus 0.08. Arm×covariate interactions, the quantity behind "this message works better for group X", show no agreement on any dataset (r from −0.57 to 0.60, centred on zero). The one output a segment-targeting tool exists to produce is the one with no measurable fidelity.

3.4 Individual fidelity: never beats a lookup table

Clarke: Sonnet 58% versus lookup 61% versus majority 60%. Evans: Sonnet MAE 3.1 g versus 2.2 g, correlation with the real person 0.01. Kerr: 0.77 versus 0.78, r=0.25. Mora-Plazas: all at the majority rate. Real covariates do not make a twin predict its person, consistent with [16, 3].

3.5 Design audit: ask them or test them

Before the twin study we ran the Clarke task two ways with the same synthetic personas and models. Poll: allocate 100 points across the six schemes by how much each "would change what you buy". Replicate: as above. Polled personas on all three Claude models put most points on calorie badges and least on graphic warnings, the reverse of the human result (population-level ρ of −0.60 for both Haiku models), with recognisably human rationales ("gory pictures would make me roll my eyes"). Under 25 rewordings the poll's modal top pick was calories-only in every segment every time. Replicated, the same personas

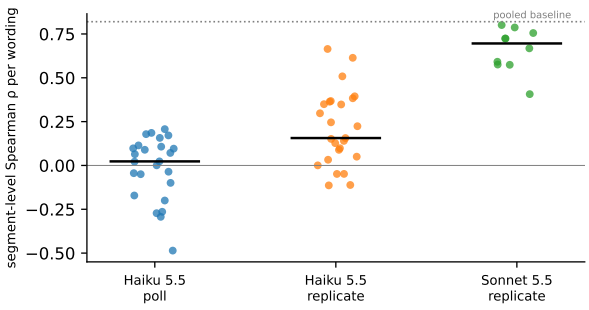


Figure 3: Segment-level ρ per prompt wording on Clarke, synthetic personas. Bars are medians.

rank pictorial warnings first. The poll is stable and wrong; replication is right on average. This is the stated-versus-revealed gap for fear appeals [18], reproduced by the simulator, and it is the design most synthetic-respondent products use.

3.6 Perturbation audit: one prompt is a sample of one

We rewrote four prompt components (persona framing, task instruction, arm descriptions, output format) five ways each and ran a 5×5 Graeco-Latin square of 25 wordings on Haiku (both designs) and 10 on Sonnet (replicate), same personas (Figure 3). Haiku's segment-level ρ spanned -0.11 to 0.66 (median 0.16); Sonnet's spanned 0.41 to 0.80 (median 0.70). The model gap survives (Mann–Whitney $p{=}10^{-5}$) but the extremes overlap. Base rate was 90% explained by the task wording on Haiku: three wordings containing an honesty admonition ("do not try to be healthy") gave 4–6% healthy choice, two without gave 24–29%; Sonnet moved from 0.22 to 0.63 around the human 0.41. Orderings were far more stable than levels (Sonnet population-level ρ never below 0.77). Every single-prompt leaderboard entry, including our own earlier 0.79 for Sonnet, is a draw from a distribution wider than most model gaps.

4 Discussion

Three fidelities give three verdicts. Effect fidelity is partly achieved by a capable model: orderings are recovered where humans have them, nothing is invented where they do not, but magnitudes are wrong by factors of five in one direction or absent in the other. Distributional fidelity is not achieved by any configuration tested. Mechanism fidelity splits exactly along the line that matters commercially: main effects of who people are come out with the right sign and several times the right strength; moderation of treatment by who people are does not come out at all.

The practical rules that follow: replicate the decision, never poll the persona; report rankings with a scale factor fitted on a small human pilot, never levels; include a null task; publish a wording family and report median and spread; and do not report segment-specific message recommendations until arm×covariate fidelity has been shown on held-out human data, which no configuration here achieves. Prompt omniscience is a separate confound: any fact placed in the stimulus text is used with certainty (Mora-Plazas, version 1), so stimulus descriptions must contain only what a human would plausibly attend to, and that choice should be perturbed like any other wording.

Limitations. One draw per twin per arm; one wording per task in the twin study (the perturbation audit covers only Clarke); Anthropic models only; stimuli are text reconstructions of visual labels; backstories were written by the model under test, so a weaker model also writes weaker twins; Kerr and Mora-Plazas outcomes are intentions and hypothetical choices rather than purchases. The datasets with published demographic heterogeneity (Hall et al. 2021; Wolfson et al. 2022) are available only on request and would be the right place to test interaction fidelity properly.

5 Conclusion

On four open behavioural trials, LLM twins built from real respondents' covariates rank message variants roughly as people do, exaggerate or miss the size of the effect, show a third to two-thirds of human variance, treat demographics as several times more deterministic than they are, and show no fidelity on which groups respond to which message. Evaluations that report one accuracy number from one prompt on survey items cannot see any of this. The three-fidelity eval, the null task and the wording family are released so that they can.

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